

- c) Jose has never received an email message from Deborah:
 $\neg (M(\text{Deborah}, \text{Jose}))$
- d) Every student in your class has sent an email message to Ken:
 $\forall x M(x, \text{Ken})$
- e) No one in your class has telephoned Nina:
 $\neg \exists x T(x, \text{Nina})$
- f) Everyone in your class has either telephoned Avi or sent him an email message:
 $\forall x (T(x, \text{Avi}) \vee M(x, \text{Avi}))$
- g) There is a student in your class who has sent everyone else in your class an email message:
 $\exists x \forall y (x \neq y \Rightarrow M(x, y))$
- h) There is someone in your class who has either sent an email message or telephoned everyone else in your class:
 $\exists x \forall y (x \neq y \Rightarrow (M(x, y) \vee T(x, y)))$
- i) There are two different students in your class who have sent each other email messages:
 $\exists x \exists y ((x \neq y) \wedge M(x, y) \wedge M(y, x) \wedge \forall z \forall w ((z \neq w) \wedge M(z, w) \wedge M(w, z) \Rightarrow ((z = x \wedge w = y) \vee (z = y \wedge w = x))))$
- j) There is a student who has sent himself or herself an email message:
 $\exists x M(x, x)$
- k) There is a student in your class who has not received an email message from anyone else in the class and who has not been called by any other student in the class:
 $\exists x \forall y (\neg M(y, x) \wedge \neg T(x, y))$
- l) Every student in the class has either received an email message or a telephone call from another student in the class:
 $\forall x \exists y (M(y, x) \vee T(y, x))$
- m) There are at least two students in your class such that one student has sent the other email and the second student has telephoned the first student:
 $\exists x \exists y ((x \neq y) \wedge M(x, y) \wedge T(y, x))$
- n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class:
 $\exists x \exists y \forall z ((x \neq y) \wedge (x \neq z) \wedge (y \neq z) \Rightarrow (M(z, x) \vee T(z, x) \vee M(z, y) \vee T(z, y)))$